
序号:



三峡大学
CHINA THREE GORGES UNIVERSITY

《可编程控制器实验》

实验报告

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日 期: 2022 年 06 月 20 日

Index

1	Experiment 1: PLC control of stage lights	1
1.1	Purpose of the experiment	1
1.2	Experimental content and steps	1
1.3	Experimental wiring (with wiring diagram)	2
1.4	Experimental program:	2
1.5	Experimental results (with photos).	4
1.6	Experiment summary:	4
2	Experiment 2: 8 LED digital tube display control	5
2.1	Purpose of the experiment	5
2.2	Experimental content and steps	5
2.3	Experimental wiring (with wiring diagram)	7
2.4	Experimental program	7
2.5	Experimental results	12
2.6	Experiment summary:	12
3	Experiment 3: Automatic control of traffic lights	13
3.1	Purpose of the experiment	13
3.2	Experiment content and steps	13
3.3	Experimental wiring	14
3.4	Experimental program	14
3.5	Experimental results	17
3.6	Experiment summary:	17
4	Experiment 4: Material Mix Control	18
4.1	Purpose of the experiment	18
4.2	Experimental content and steps	18
4.3	Experimental wiring	19
4.4	Experimental program	19
4.5	Experimental results	21
4.6	Experiment summary:	21
5	Experiment 5: Control of the four-stage conveyor belt	21
5.1	Purpose of the experiment	21
5.2	Experimental content and steps	21
5.3	Experimental wiring	23

5.4	Experimental program	24
5.5	Experimental results.....	30
5.6	Experiment summary.....	30
6	Experiment 6: Water tower water level control	30
6.1	Purpose of the experiment.....	30
6.2	Experimental content and steps.....	30
6.3	Experimental wiring.....	32
6.4	Experimental program	33
6.5	Experimental results.....	36
6.6	Experiment summary.....	36

1 Experiment 1: PLC control of stage lights

1.1 Purpose of the experiment

- ✓ 1. Master the application of shift registers instructions
- ✓ 2. Master the interface between PLC and peripheral circuits

1.2 Experimental content and steps

1. Design requirements

Stage lighting control can be controlled by PLC, such as lighting, shifting and various timing changes. The stage light control module consists of 7 sets of indicators, as shown below:

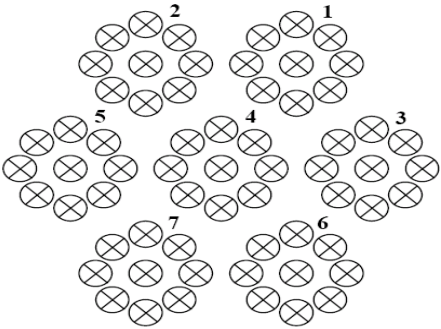


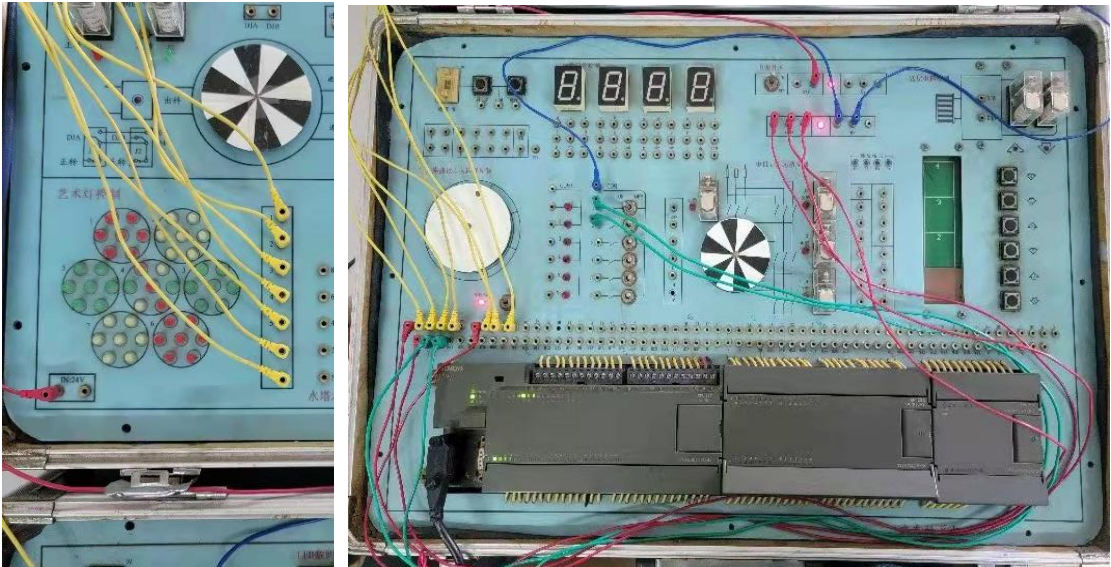
Fig1: Stage light control schematic

The timing for the 1 to 7 sets of lamps to be flashed is as follows:

1. The lamps 1 to 7 are lit in turn, and then fully lit.
2. Repeat (1), cycle back and forth.
3. Determine the input and output ports and write the program
4. Compile the program, download it to the memory of the PLC host after error-free, and run the program.
5. Debug the program until it meets the design requirements.

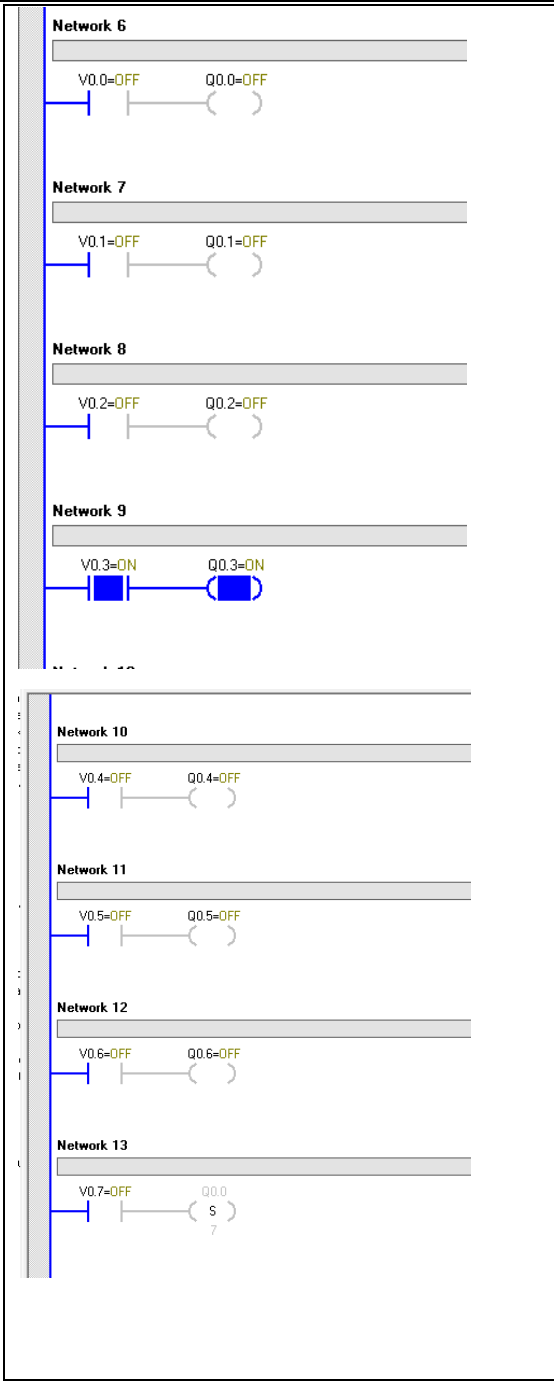
Input			Output		
Host	Module	Comment	Host	Module	Comment
M00	K1	Start	00	1	Light 1
M01	K2	Stop	01	2	Light 2
			02	3	Light 3
1M	24V		03	4	Light 4
COM	0V	Switch common	04	5	Light 5
			05	6	Light 6
			06	7	Light 7
			1L	24V	
			2L	24V	

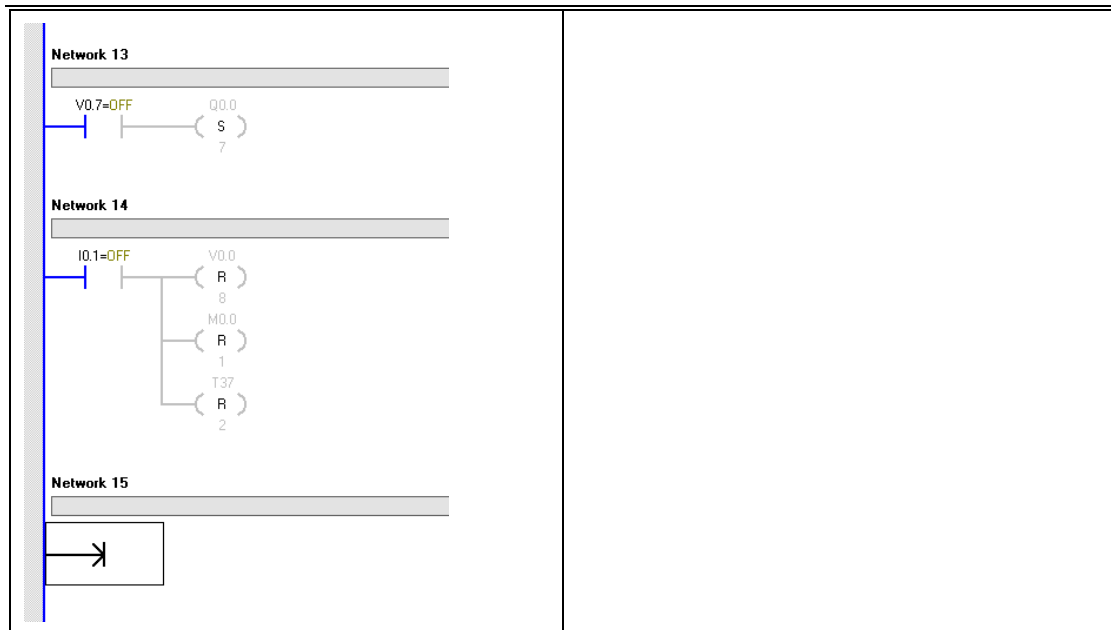
1.3 Experimental wiring (with wiring diagram)



1.4 Experimental program:

LADDER	STL
PROGRAM COMMENTS Network 1 Network Title Network Comment I0.0=ON I0.1=OFF M0.0=ON M0.0=ON Network 2 M0.0=ON T34=OFF +S9=T33 50 PT 10 ms Network 3 T33=ON +S9=T34 50 PT 10 ms Q1.7=ON Network 3 T33=OFF +S0=T34 50 PT 10 ms Q1.7=OFF Network 4 V0.7=OFF N M0.0=ON P MOV_B ENO 16#1 IN VB0-VB0 Network 5 Q1.7=OFF P ROL_B ENO VB0-VB0 1-N VB0-VB0	PROGRAM COMMENTS Network 1 Network Title Network Comment LD I0.0 O M0.0 AN I0.1 = M0.0 Network 2 LD M0.0 AN T34 TON T33, 50 Network 3 LD T33 TON T34, 50 = Q1.7 Network 4 LD V0.7 ED LD M0.0 EU OLD MOVB 16#1, VB0 Network 5 LD Q1.7

	EU RLB VB0, 1 Network 6 LD V0.0 = Q0.0 1 / 4 task1 / MAIN (OB1) Network 7 LD V0.1 = Q0.1 Network 8 LD V0.2 = Q0.2 Network 9 LD V0.3 = Q0.3 Network 10 LD V0.4 = Q0.4 Network 11 LD V0.5 = Q0.5 Network 12 LD V0.6 = Q0.6 Network 13 LD V0.7 S Q0.0, 7 Network 14 LD I0.1 R V0.0, 8 R M0.0, 1 R T37, 2
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1.5 Experimental results (with photos)



1.6 Experiment summary:

This is our first project in this course. In this project we learn how to code in plc in ladder language. And how to upload the program in the plc device

In this project we learn about how to control the street light. This is a basic program for beginner.in this project our honorable laoshi Lina laoshi help us a lot.

2 Experiment 2: 8 LED digital tube display control

2.1 Purpose of the experiment

1. Master the application of shift registers instructions
2. Master the use of PLC to control digital tube display

2.2 Experimental content and steps

1. PLC 实验箱一台(PLC Experiment box)
2. LED 数码管控制模块(LED digital tube)
3. 编程电缆一根(program cable)
4. 导线若干(wire)

实验内容及步骤

1、设计要求

Design a digital tube cycle display program. The value of the value is displayed from 0 to 9. The digital tube is of the common cathode type. A, B, C, D, E, F, G, Dp are digital pipe segment codes, COM is the digital pipe common terminal (bit code), when the segment code is input to the high level, the bit code is input to the low level, the corresponding segment point bright.

2、determine the input and output ports, and write a program

3、Compile the program, download it to the memory of the PLC host after error-free, and run the program.

4、debug the program until it meets the design requirements

Input			Output		
Host	Experimental module	Comment	Host	Experimental module	Comment
I0.0	Starter	start up	Q0.0	A	Segment code A
I0.1	Stop	stop	Q0.1	B	Segment code B
I0.2	K1	0-9 loop	Q0.2	C	Segment code C
I0.3	K2	Segment loop	Q0.3	D	Segment code D
			Q0.4	E	Segment code E
			Q0.5	F	Segment code F
			Q0.6	G	Segment code G
			Q0.7	DP	Decimal point
	COM←→0V	COM		COM←→0V	LED COM
1M	24V		1L	24V	
			2L	24V	

交通信号灯的自动控制

实验目的

- 1、掌握 PLC 功能指令的用法
- 2、掌握用 PLC 控制交通灯的方法

实验器材

- 1、PC 机一台
- 2、PLC 实验箱一台
- 3、交通灯控制模块
- 4、编程电缆一根
- 5、导线若干

实验内容及步骤

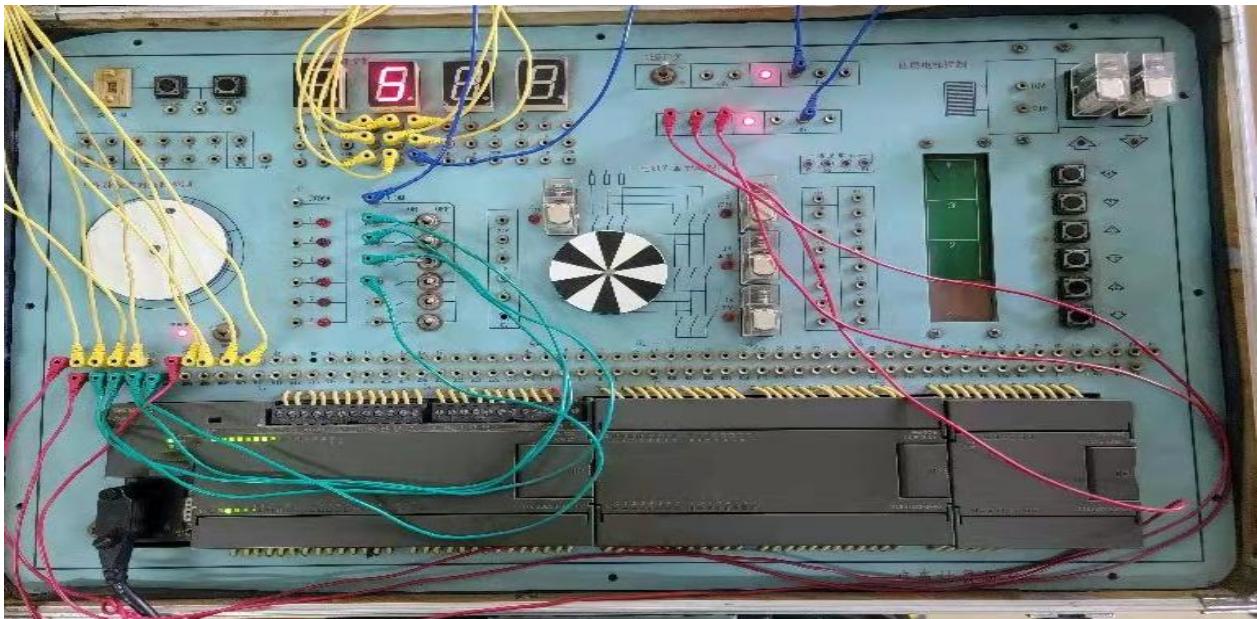
1、设计要求

Design a control program for traffic lights at intersections. The requirements are: north and south to red light for 10 seconds, east and west green light for 4 seconds, flash for 3 seconds, east and west to yellow light for 3 seconds, then east and west to red light for 10 seconds, north and south to green light for 4 seconds, flash for 3 seconds, north and south to yellow The light is on for 3 seconds and is continuously looped。

- 2、确定输入、输出端口、并编写程序。
- 3、编译程序，无误后下载至 PLC 主机的存储器中，并运行程序。
- 4、调试程序，直至符合设计要求。
- 5、参考程序（s7_226_traff.mwp）及接线表

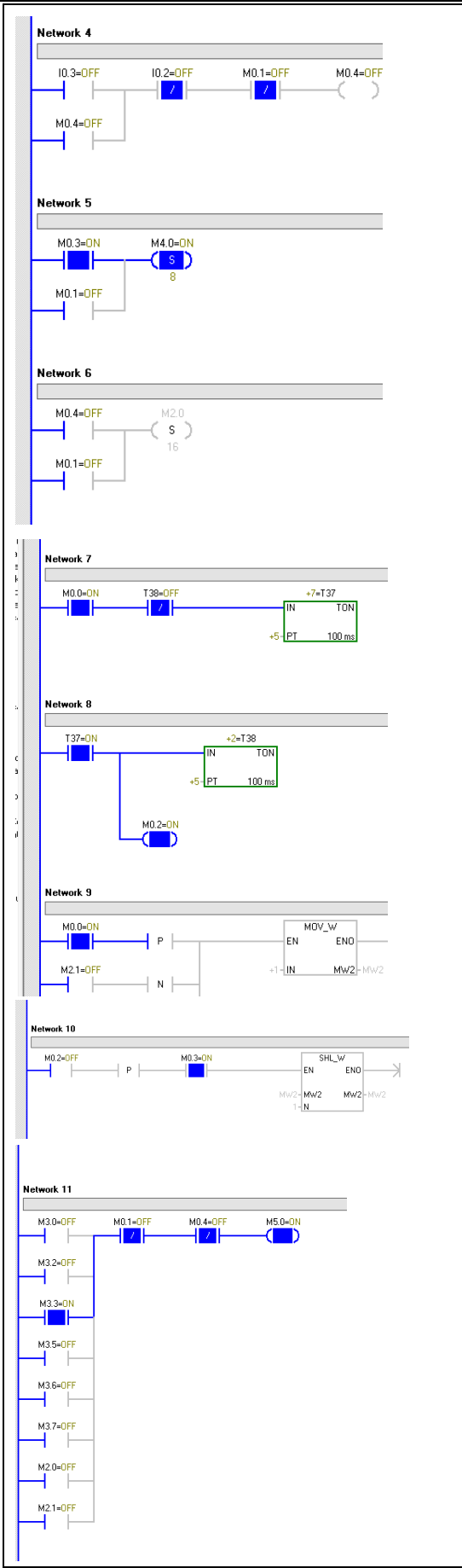
输入			输出		
主机	实验模块	注释	主机	实验模块	注释
M00	K1	启动	00	A1	红灯（上下）
M01	K2	停止	01	A2	绿灯（上下）
			02	A3	黄灯（上下）
			03	B1	红灯（左右）
			04	B2	绿灯（左右）
			05	B3	黄灯（左右）
			CA1	B1	人行道红灯（上下）
			CA2	B2	人行道绿灯（上下）
			CB1	A1	人行道红灯（左右）
			CB2	A2	人行道绿灯（左右）
1M	24V		1L	24V	
COM	0V	开关公共端			

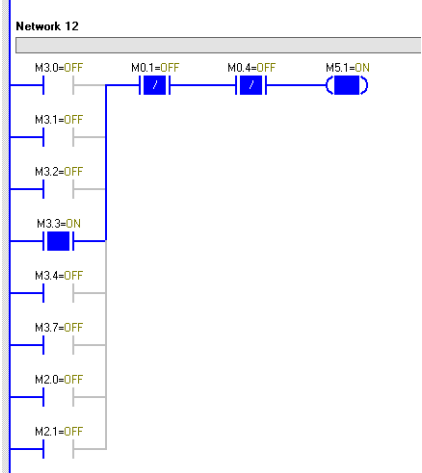
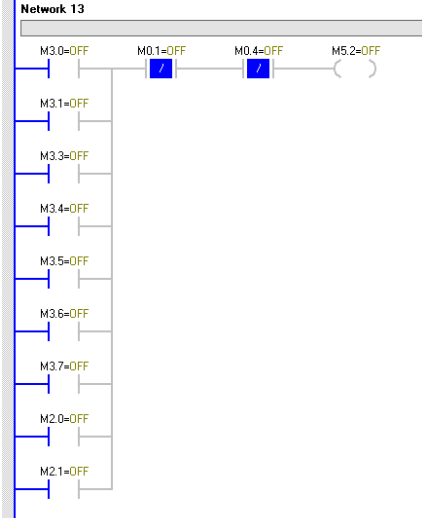
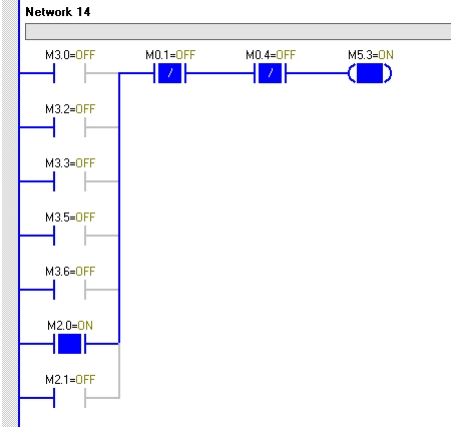
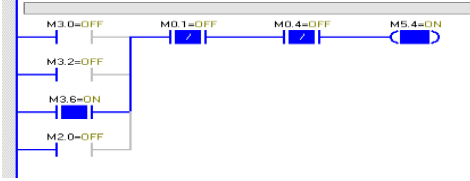
2.3 Experimental wiring (with wiring diagram)

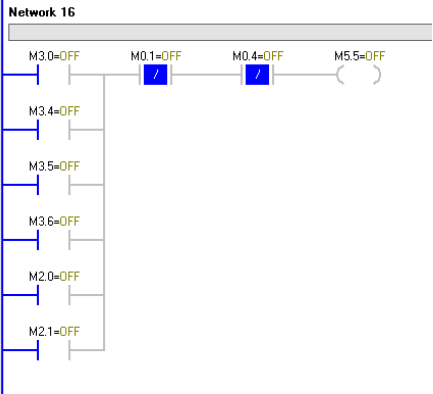
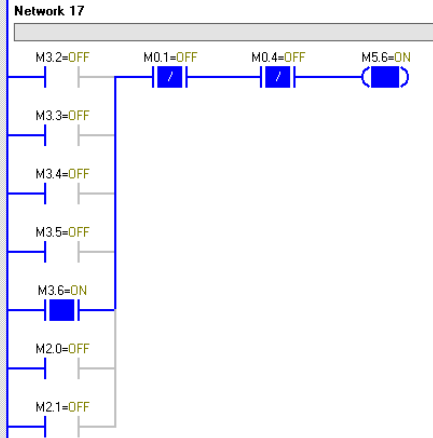
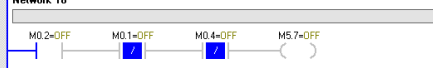
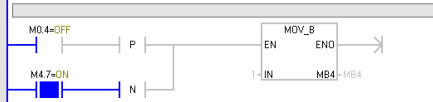
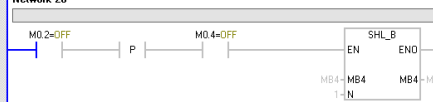


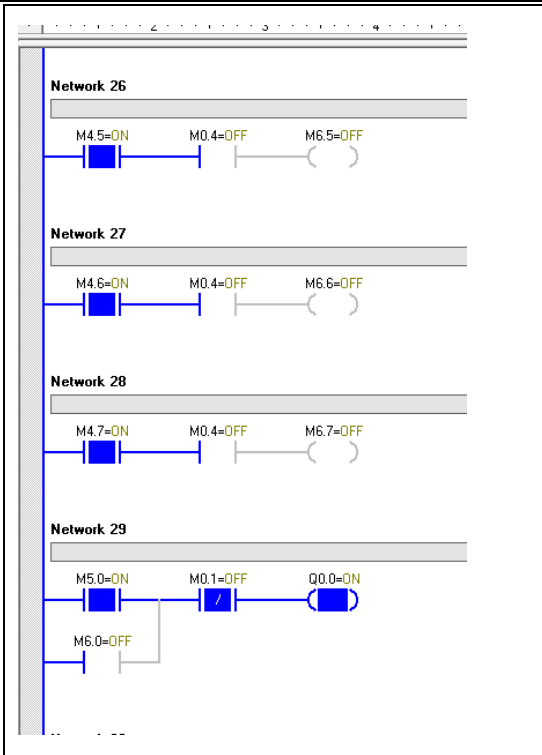
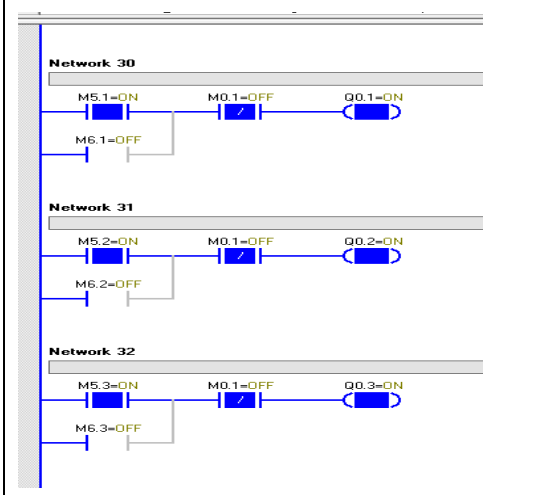
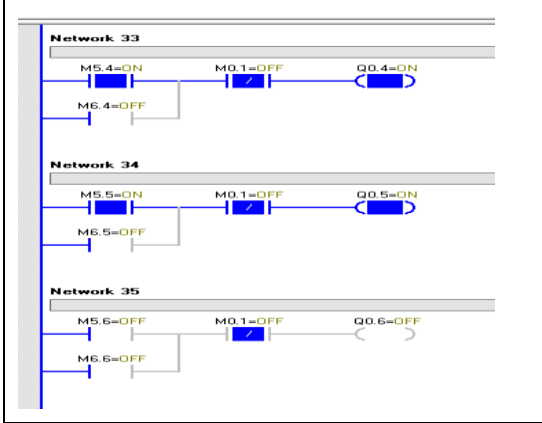
2.4 Experimental program

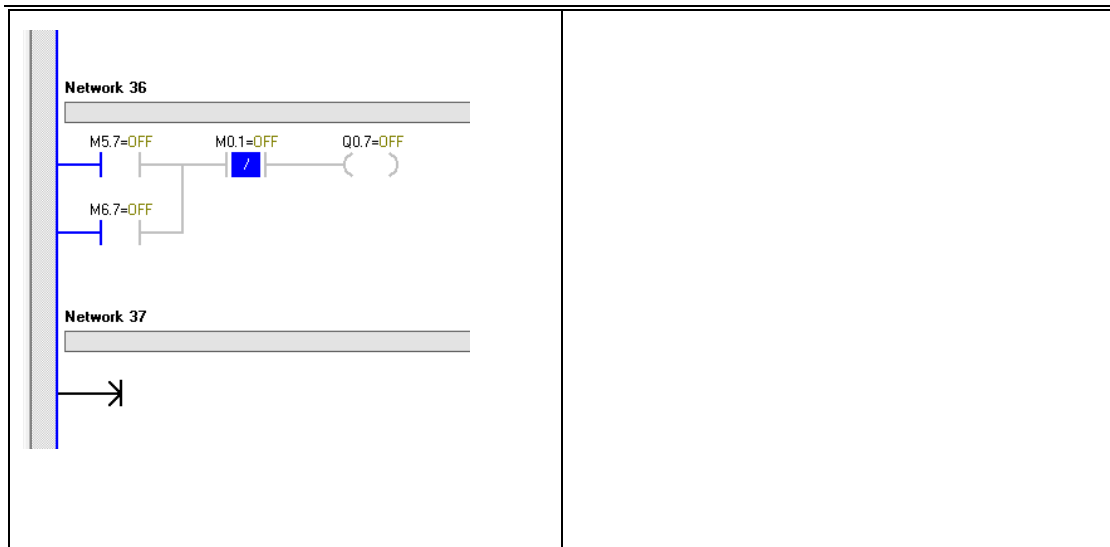
LADDER	STL
Project 3	PROGRAM COMMENTS Network 1 Network Title Network Comment LD I0.0 O M0.0 AN I0.1 = M0.0 Network 2 LD M0.0 AN I0.1 EU

	LD T42 ED OLD TOF T37, 40 Network 3 LD T37 = M10.0 Network 4 LD T37 ED TOF T38, 30 Network 5 LD T38 = M10.1 Network 6 LD T38 ED TOF T40, 30 1 / 5 task3 / MAIN (OB1) Network 7 LD T39
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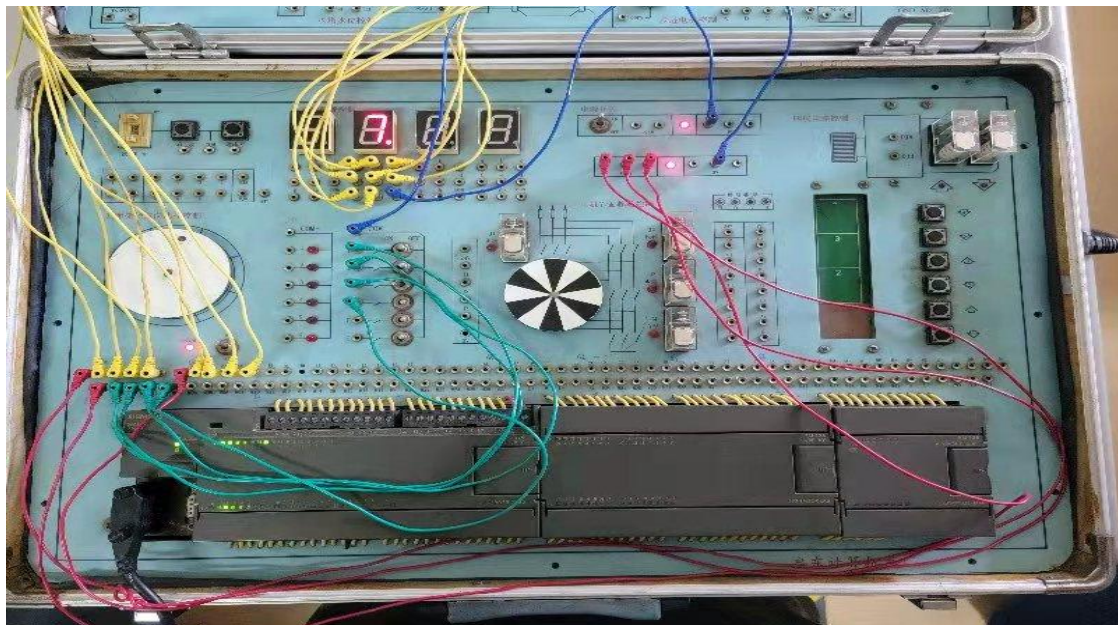
	= M10.2
	Network 8
	LD T39
	ED
	TOF T40, 40
	Network 9
	LD T40
	= M10.3
	Network 10
	LD T40
	ED
	TOF T41, 30
	Network 11
	LD T41
	= M10.4
	Network 12
	LD T41
	ED
	TOF T42, 30
	Network 13
	LD T42
	= M10.5

	Network 14
	LD M10.0
	O M10.1
	O M10.2
	= Q0.0
	Network 15
	LD M10.0
	O M10.1
	= Q0.4
	2 / 5
	task3 / MAIN (OB1)
	Network 16
	LD M10.2
	A SM0.5
	= Q0.5
	Network 17
	LD M10.3
	O M10.4
	O M10.5
	= Q0.3
	Network 18
	LD M10.3

	O M10.4 = Q0.1 Network 19 LD M10.5 A SM0.5 = Q0.2 Network 20 LD I0.1 R T37, 42
	R M10.0, 6 R M0.0, 1
	



2.5 Experimental results



2.6 Experiment summary:

This is our second project in this course in this project we learn about the basic principle of a seven-segment display and how to control a 7 seg display with a plc. by using ladder diagram we write down a program to display 1 to 9 in 7 seg display. We also learn about the basic wiring diagram of PLC. We also learn about the input output terminal of the PLC.

3 Experiment 3: Automatic control of traffic lights

3.1 Purpose of the experiment

1. Master the usage of PLC function instructions
2. Master the method of controlling traffic lights with PLC

实验器材

- 1、PC 机一台
- 2、PLC 实验箱一台
- 3、交通灯控制模块
- 4、编程电缆一根
- 5、导线若干

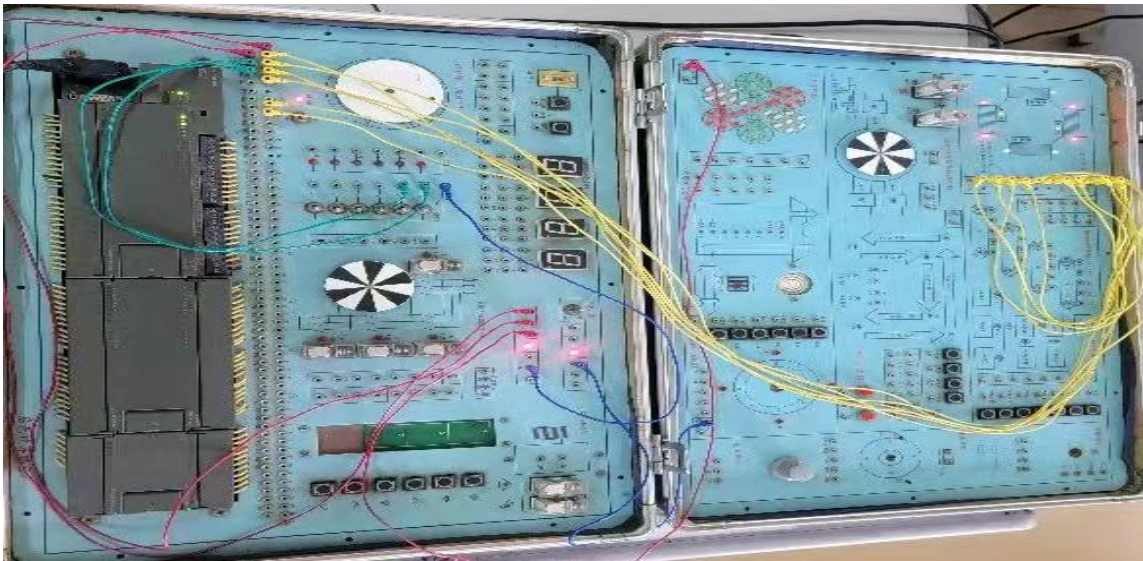
3.2 Experiment content and steps

Design requirements

- Design a control program for traffic lights at intersections. The requirements are: north and south to red light for 10 seconds, east and west green light for 4 seconds, flash for 3 seconds, east and west to yellow light for 3 seconds, then east and west to red light for 10 seconds, north and south to green light for 4 seconds, flash for 3 seconds, north and south to yellow the light is on for 3 seconds and it repeats repeatedly.
2. Determine the input and output ports and write the program.
 3. Compile the program, download it to the memory of the PLC host after error-free, and run the program.
 4. Debug the program until it meets the design requirements.

Input			Output		
HOST	实验模块	COMMENT	HOST	实验模块	COMMENT
M00	K1	STARTER	00	A1	Red light (upper and lower)
M01	K2	STOP	01	A2	Green light (up and down)
			02	A3	Yellow light (upper and lower)
			03	B1	Red light (left and right)
			04	B2	Green light (left and right)
			05	B3	Yellow light (left and right)
			CA1	B1	Sidewalk red light (up and down)
			CA2	B2	Sidewalk green light (up and down)
			CB1	A1	Sidewalk red light (left and right)
			CB2	A2	Sidewalk green light (left and right)
1M	24V		1L	24V	
COM	0V	开关公共端			

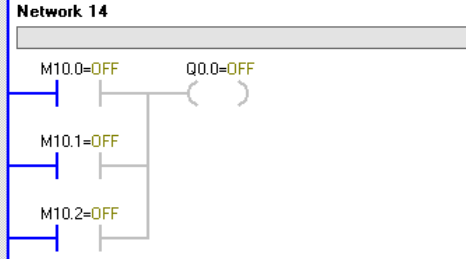
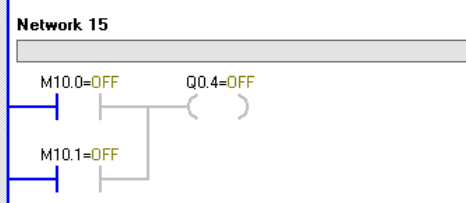
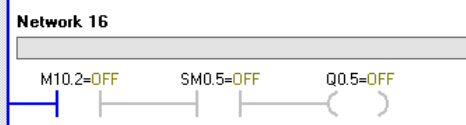
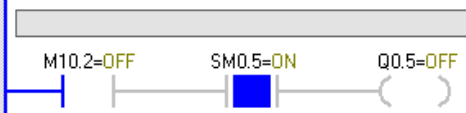
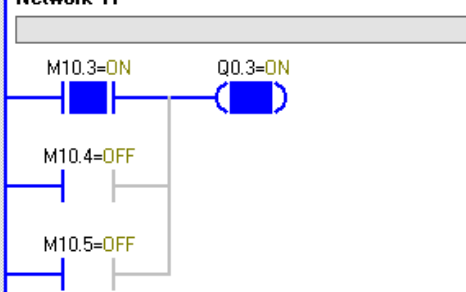
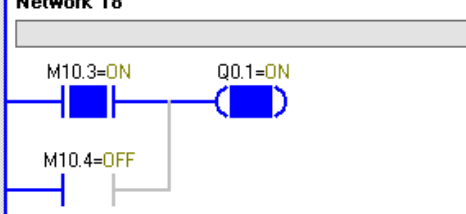
3.3 Experimental wiring

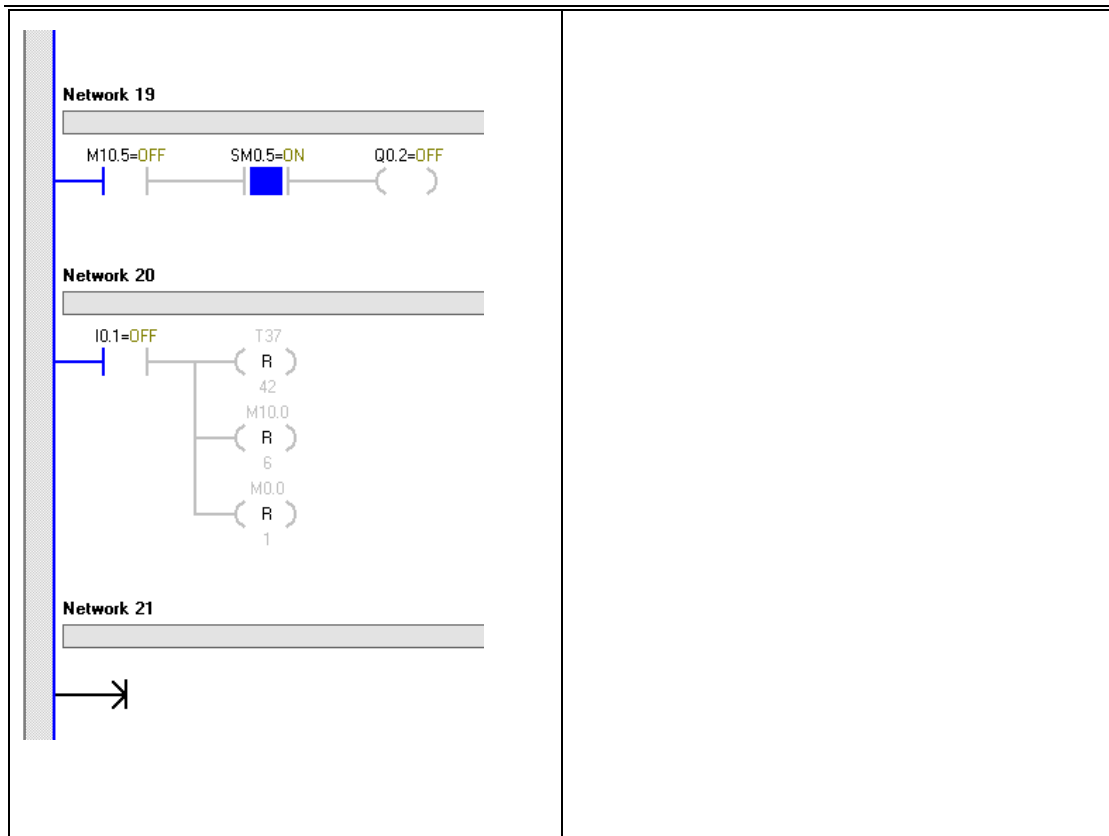


3.4 Experimental program

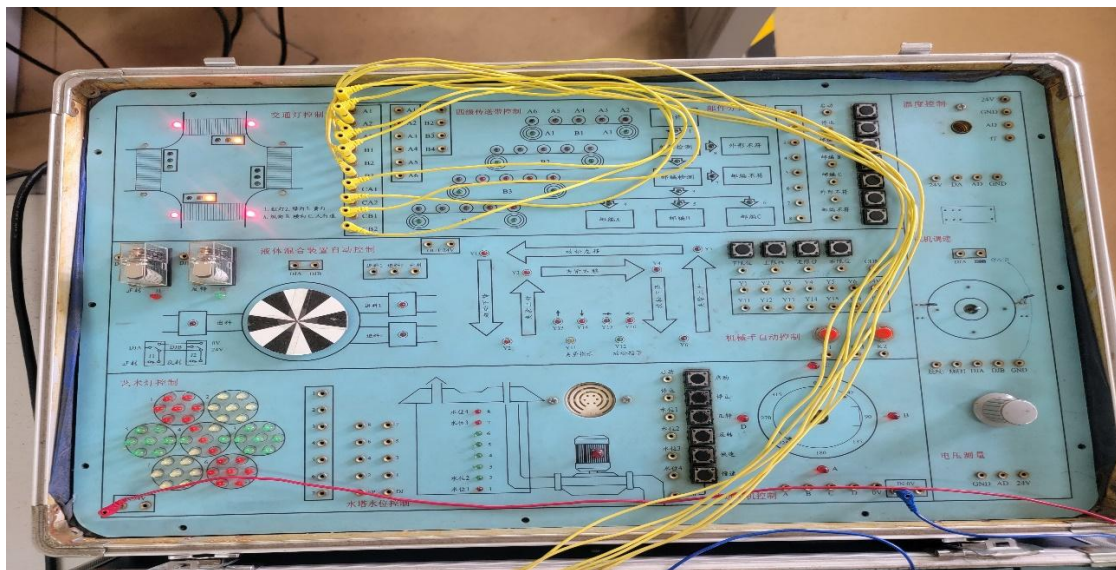
LADDER	STL
	PROGRAM COMMENTS Network 1 Network Title Network Comment LD I0.0 O M0.0 AN I0.1 = M0.0 Network 2 LD M0.0 AN I0.1 EU LD T42 ED OLD TOF T37, 40 Network 3 LD T37 = M10.0 Network 4 LD T37 ED TOF T38, 30 Network 5 LD T38

	= M10.1 Network 6 LD T38 ED TOF T40, 30 1 / 5 task3 / MAIN (OB1) Network 7 LD T39 = M10.2 Network 8 LD T39 ED TOF T40, 40 Network 9 LD T40 = M10.3 Network 10 LD T40 ED TOF T41, 30 Network 11 LD T41 = M10.4 Network 12 LD T41 ED TOF T42, 30 Network 13 LD T42 = M10.5 Network 14 LD M10.0 O M10.1 O M10.2 = Q0.0 Network 15 LD M10.0 O M10.1 = Q0.4 2 / 5 task3 / MAIN (OB1) Network 16 LD M10.2
--	---

Network 14  Network 15  Network 16 	A SM0.5 = Q0.5 Network 17 LD M10.3 O M10.4 O M10.5 = Q0.3 Network 18 LD M10.3 O M10.4 = Q0.1 Network 19 LD M10.5 A SM0.5 = Q0.2 Network 20 LD I0.1 R T37, 42 R M10.0, 6 R M0.0, 1
Network 16  Network 17  Network 18 	



3.5 Experimental results



3.6 Experiment summary:

This is our 3rd project in this course in this project we learn about how traffic light work and its working principle. We write down a program in ladder diagram to control the traffic light with the help of step7 PLC. In this project we learn a lot of commands like MOV command.

4 Experiment 4: Material Mix Control

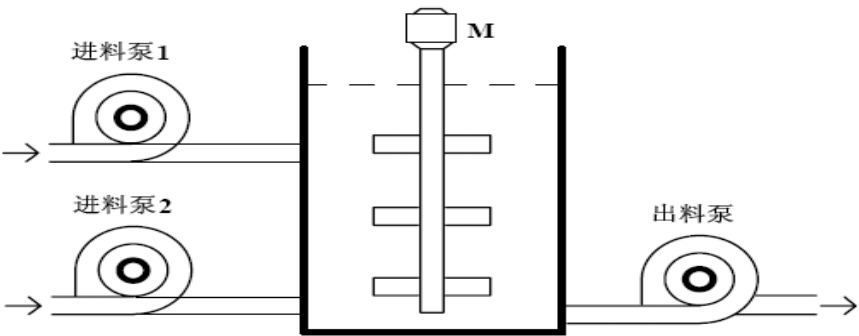
4.1 Purpose of the experiment

- 1.master the use of functional instructions
- 2.master the design of material mixing control program

4.2 Experimental content and steps

Design requirements

Design a material mixing control program, the module diagram is as follows:



Material mixing system schematic

It is required to start the feed pump 1 first. After the feed is completed, the feed pump 1 is closed, and then the feed pump 2 is started. After the feed is completed, the feed pump 2 is closed, the mixer starts to work, and the forward rotation is performed for 5 minutes, then reversed. After 5 minutes, the mixer stopped working and the discharge valve was opened. After the preparation is completed, the above process is repeated. When the stop button is pressed, if it is in a loop, then wait for the current loop to end, that is, after the discharge is completed, the program stops running.

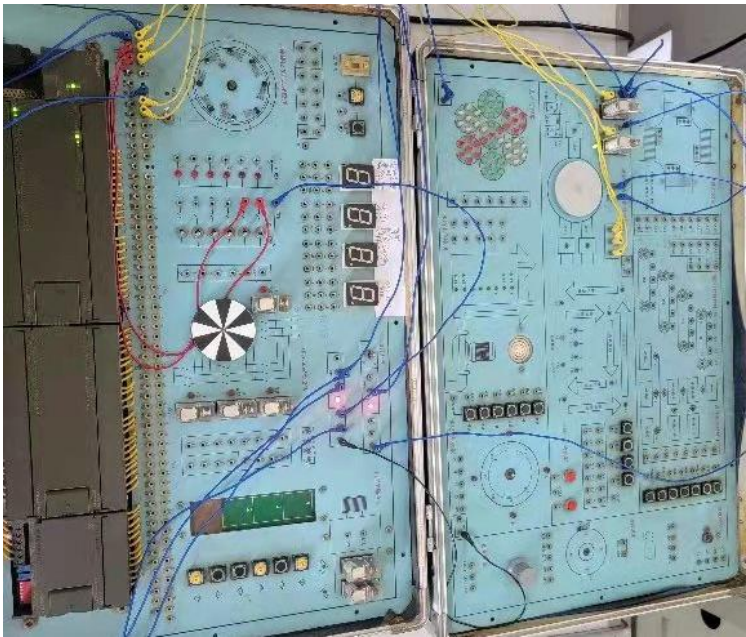
2.determine the input and output ports, and write a program

3.Compile the program, download it to the memory of the PLC host after error-free, and run the program.

4.debug the program until it meets the design requirements.

Input			Output		
Host	Experimental module	comment	Host	Experimental module	comment
I0.0	Start	Start	Q0.0	Feed 1	Q0.0
I0.1	Stop	Stop	Q0.1	Feed 2	Q0.1
			Q0.2	Discharge	Q0.2
1M	24V		1L	24V	
	0V←→COM		Q0.4	Forward	Refer to the wiring diagram on the left side of the motor for relay connection.
			Q0.5	Reverse	
			2L	0V	

4.3 Experimental wiring

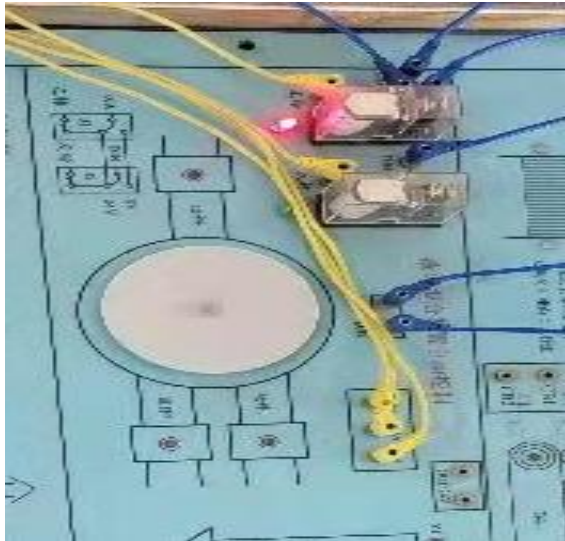


4.4 Experimental program

LADDER	STL
PROGRAM COMMENTS Network 1 Network Title Network Comment I0.0=ON I0.1=OFF M0.0=ON M0.0=ON Network 2 I0.1=OFF I0.0=ON M0.1=OFF M0.1=OFF Network 3 M0.1=OFF T 37 R 6 Network 4 M0.0=ON P +30=T 37 IN TOF 30 PT 100 ms T42=OFF N	PROGRAM COMMENTS Network 1 Network Title Network Comment LD I0.0 O M0.0 AN I0.1 = M0.0 Network 2 LD I0.1 O M0.1 AN I0.0 = M0.1 Network 3 LD M0.1 R T37, 6 Network 4 LD M0.0 EU LD T42 ED OLD TOF T37, 30 Network 5

Network 5 T37=OFF Q0.0=OFF Network 6 T37=OFF N +30=T38 TOF 30 PT 100 ms Network 7 T38=OFF Q0.1=OFF Network 8 T38=OFF N +100=T39 TOF 100 PT 100 ms Network 9 T39=OFF Q0.5=OFF Q0.4=OFF Network 10 T39=OFF N +100=T40 TOF 100 PT 100 ms Network 11 T40=OFF Q0.4=OFF Q0.5=OFF Network 12 T40=OFF N +30=T41 TOF 30 PT 100 ms Network 13 T41=OFF Q0.2=OFF Network 14 T41=OFF N +10=T42 TOF 10 PT 100 ms Network 15 T42=OFF M0.2=OFF Network 16 End of program	LD T37 = Q0.0 Network 6 LD T37 ED TOF T38, 30 1 / 4 task4 / MAIN (OB1) Network 7 LD T38 = Q0.1 Network 8 LD T38 ED TOF T39, 100 Network 9 LD T39 AN Q0.5 = Q0.4 Network 10 LD T39 ED TOF T40, 100 Network 11 LD T40 AN Q0.4 = Q0.5 Network 12 LD T40 ED TOF T41, 30 Network 13 LD T41 = Q0.2 Network 14 LD T41 ED TOF T42, 10 Network 15 LD T42 = M0.2
--	---

4.5 Experimental results



4.6 Experiment summary:

This is our 4th project. In this project we learn about use of functional instructions and master the design of material mixing control program. This is a very important project in practical life because in practical life we have to control this kind of machine. This is a very basic project in most of the industry.

5 Experiment 5: Control of the four-stage conveyor belt

5.1 Purpose of the experiment

- 1, master the use of functional instructions
2. Master the design of the four-stage conveyor control program

实验器材

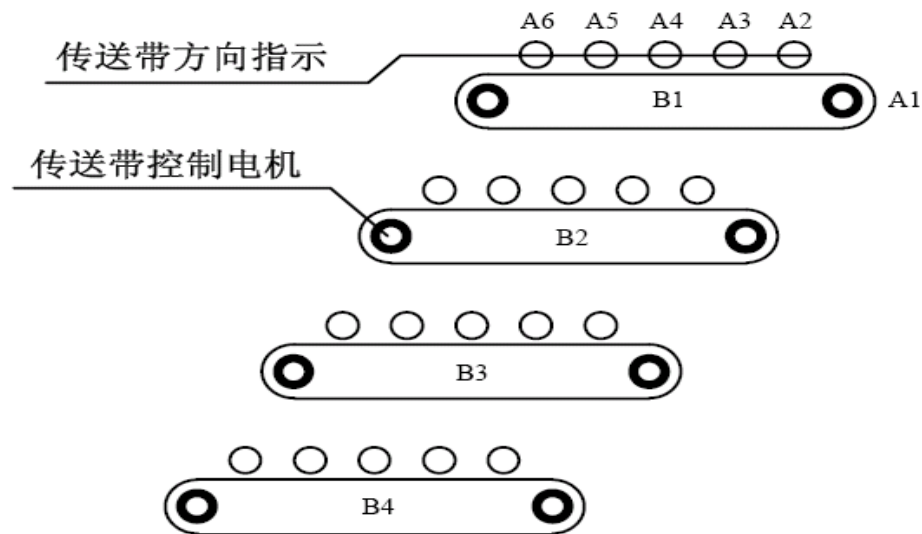
- 1、PC 机一台
- 2、PLC 实验箱一台
- 3、四级传送带控制模块
- 4、编程电缆一根
- 5、导线若干

5.2 Experimental content and steps

1. Design requirements

Design a four-stage conveyor control program.

The module diagram is as follows:



Four-stage conveyor belt schematic

The workflow of the experimental program:

The control module is a one-, two-, three-, and four-stage conveyor belt from top to bottom.

When the start button is pressed, the feed indicator lights, the four-stage conveyor starts to work, the green light is on, the red light starts to cycle from right to left; after 5 seconds delay, the three-stage conveyor starts working; after 5 seconds delay The secondary conveyor starts to work; after 5 seconds delay, the primary conveyor starts working.

Switches K1, K2, K3, and K4 are used to simulate the fault signals of the first, second, third, and fourth stages of the conveyor belt, respectively.

When K1 is closed, the primary conveyor belt fails, the primary conveyor belt stops immediately. After 5 seconds delay, the secondary conveyor belt stops. After 5 seconds delay, the tertiary conveyor belt stops. After 5 seconds delay, the 4th conveyor belt stops.

When K2 is closed, the secondary conveyor belt fails, and the primary and secondary conveyor belts stop immediately. After a delay of 5 seconds, the tertiary conveyor belt stops. After a delay of 5 seconds, the four-stage conveyor belt stops.

When K3 is closed, the third-stage conveyor belt fails, and the first-, second-, and third-stage conveyors stop immediately. After 5 seconds delay, the four-stage conveyor stops.

When K4 is closed, the four-stage conveyor belt fails, and the first, second, third and fourth conveyor belts stop immediately.

When the fault is removed, the switch is turned off and the four-stage conveyor starts working at the same time.

When the stop button is pressed, the primary conveyor stops immediately. After a delay of 5 seconds, the secondary conveyor stops. After a delay of 5 seconds, the three-stage conveyor stops. After a delay of 5 seconds, the four-stage conveyor stops.

2, determine the input and output ports, and write a program

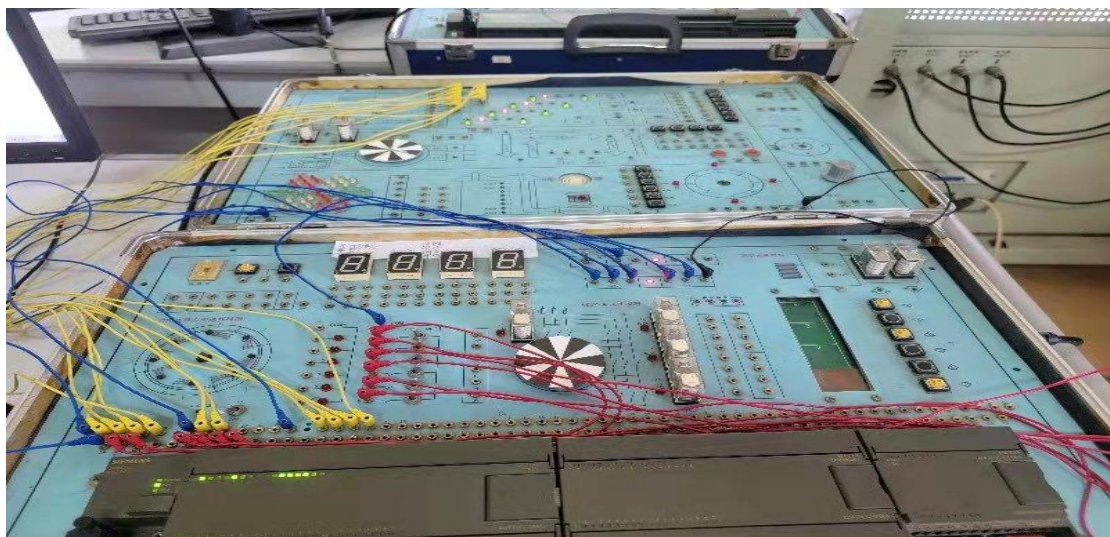
3. Compile the program, download it to the memory of the PLC host after error-free, and run the program.

4, debug the program until it meets the design requirements.

Wiring table

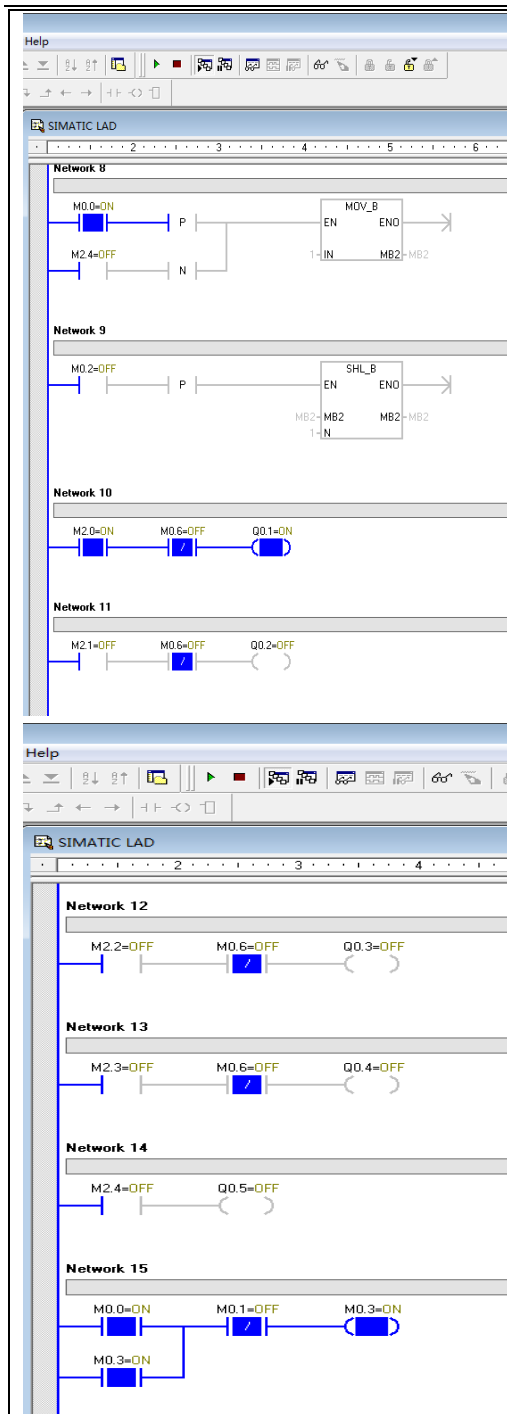
Input			Output		
Host	Experimental module	Comment	Host	Experimental module	Comment
I0.0	Start	Start	Q0.0	A1	
I0.1	Stop	Stop			
			Q0.1	A2	Belt conveying direction indication
			Q0.2	A3	
I0.3	K1	Primary conveyor failure simulation	Q0.3	A4	
I0.4	K2	Secondary conveyor failure simulation	Q0.4	A5	
I0.5	K3	Three-stage conveyor belt fault simulation	Q0.5	A6	
I0.6	K4	Four-stage conveyor belt fault simulation			
			Q1.1	B1	Primary conveyor motor
			Q1.2	B2	Secondary conveyor motor
			Q1.3	B3	Three-stage conveyor motor
			Q1.4	B4	Four-stage conveyor motor
	COM	24V	1L		24V
			2L		24V
1M		0	3L		0V

5.3 Experimental wiring

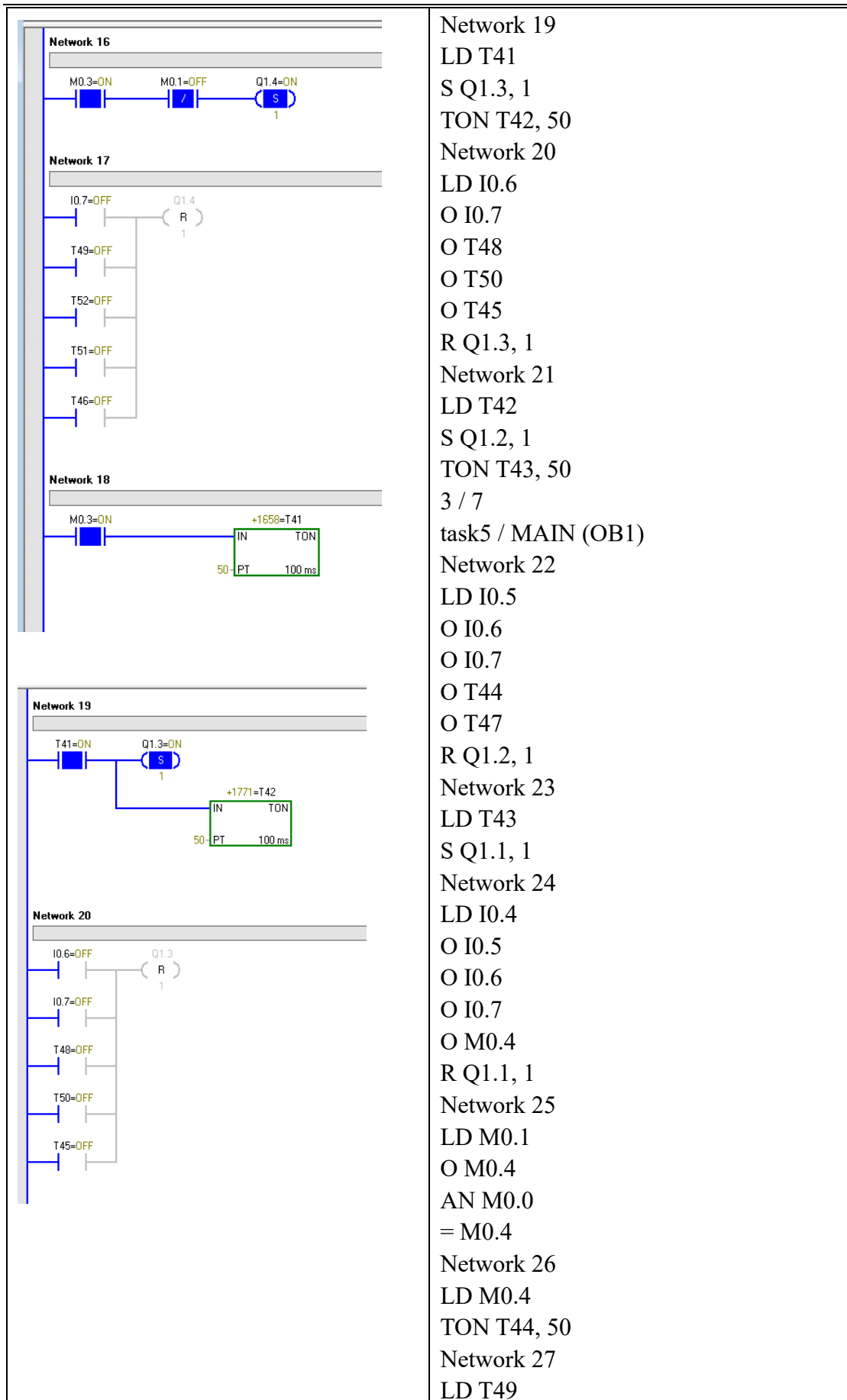


5.4 Experimental program

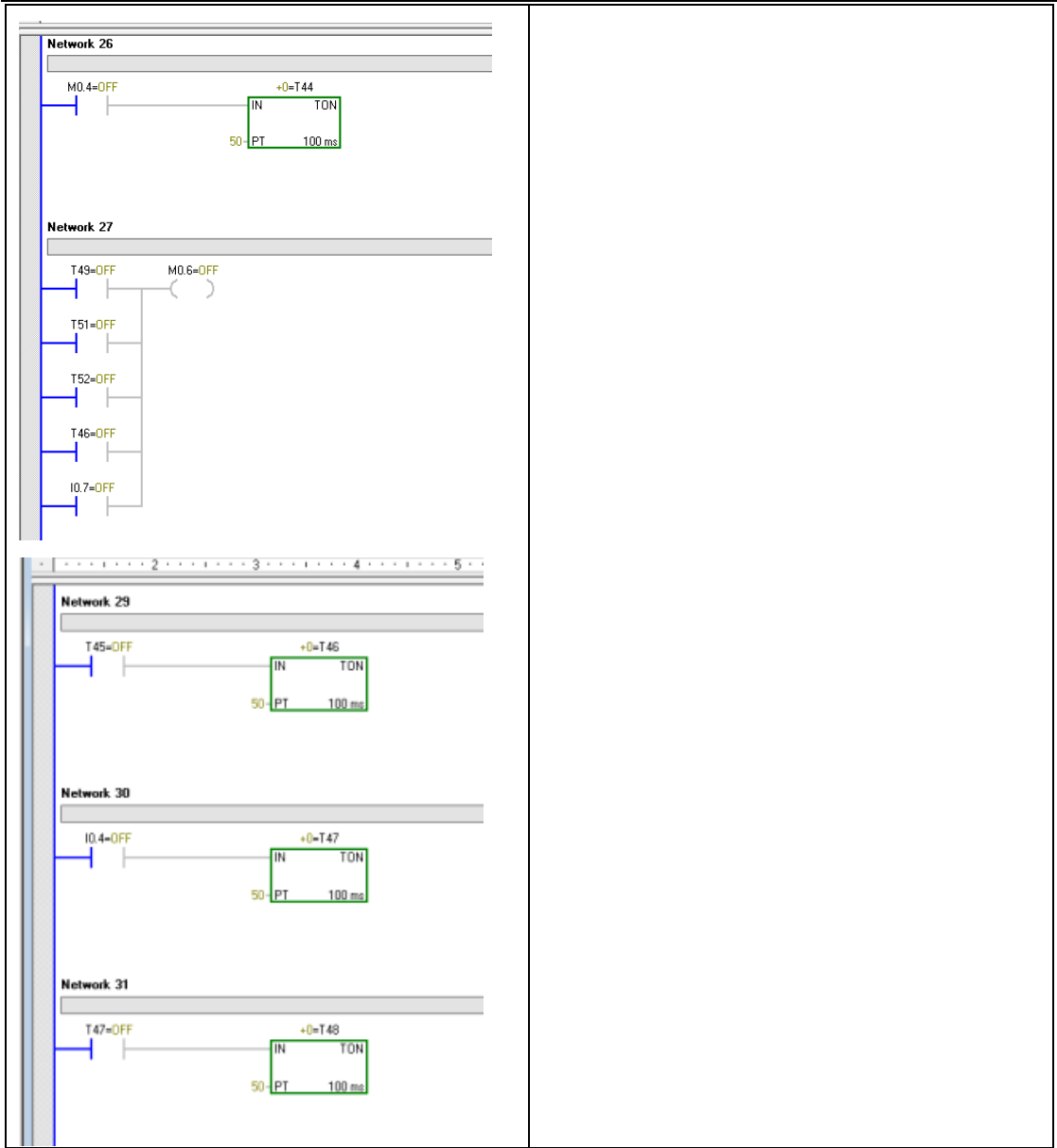
LADDER	STL
PROGRAM COMMENTS Network 1 Network Title Network Comment LD I0.0=ON O M0.0 AN I0.1=OFF = M0.0 Network 2 LD I0.1=OFF O M0.1 AN I0.0=ON = M0.1 Network 3 LD M0.0=ON S Q0.0=ON 1 Network 4 LD M0.6=OFF R Q0.0 1 Network 5 LD M10.0=ON O M10.0 AN M0.6=ON = M10.0 Network 6 LD M10.0=ON AN T39=OFF TON T40, 2 PT 100 ms Network 7 LD T40=ON TON T39, 2 PT 100 ms M10.0=ON	PROGRAM COMMENTS Network 1 Network Title Network Comment LD I0.0 O M0.0 AN I0.1 = M0.0 Network 2 LD I0.1 O M0.1 AN I0.0 = M0.1 Network 3 LD M0.0 S Q0.0, 1 Network 4 LD M0.6 R Q0.0, 1 Network 5 LD M0.0 O M10.0 AN M0.6 = M10.0 Network 6 LD M10.0 AN T39 TON T40, 2 1 / 7 task5 / MAIN (OB1) Network 7 LD T40 TON T39, 2 = M0.2 Network 8 LD M0.0 EU LD M2.4 ED OLD MOVB 1, MB2

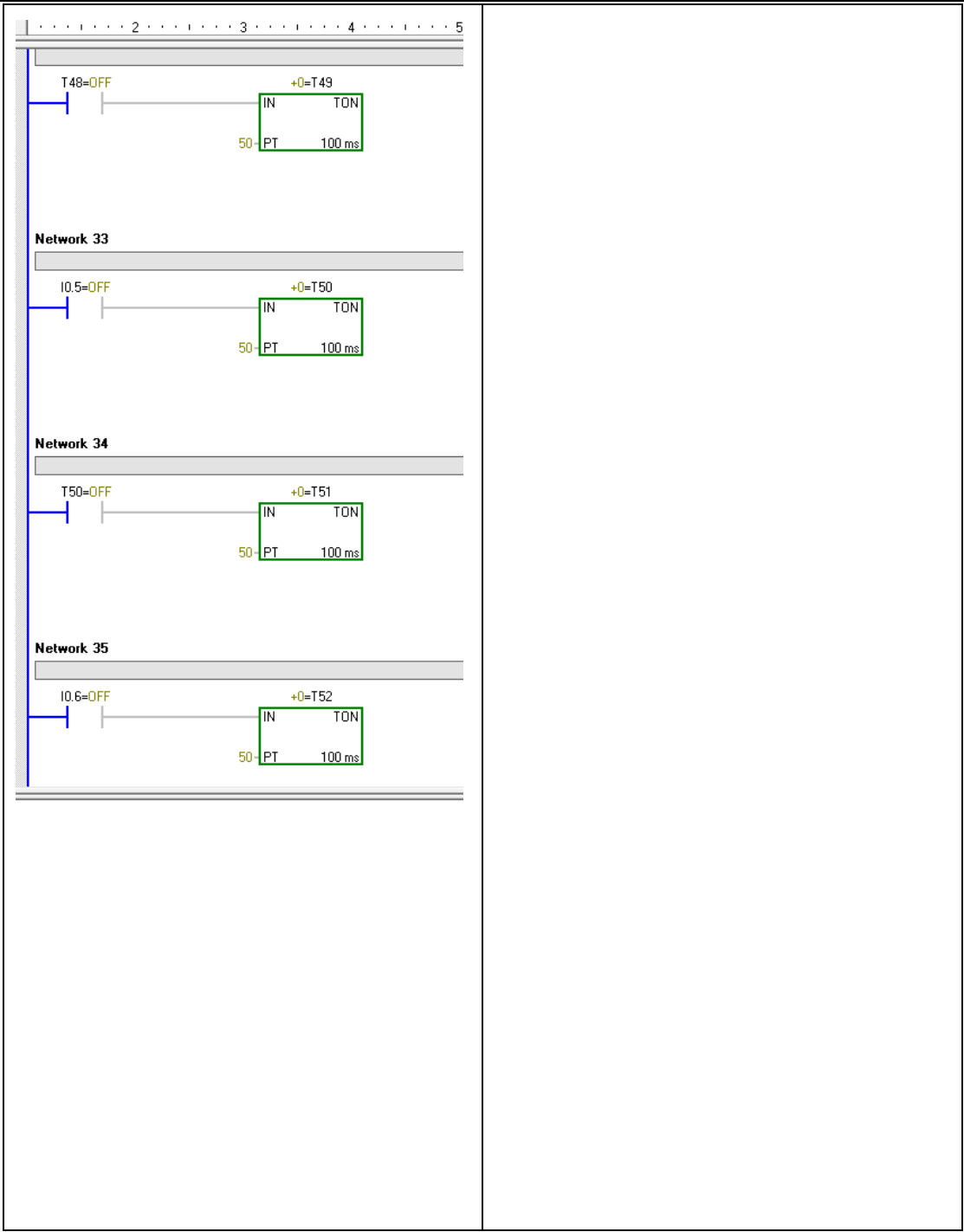


Network 9
 LD M0.2
 EU
 SLB MB2, 1
 Network 10
 LD M2.0
 AN M0.6
 = Q0.1
 Network 11
 LD M2.1
 AN M0.6
 = Q0.2
 Network 12
 LD M2.2
 AN M0.6
 = Q0.3
 Network 13
 LD M2.3
 AN M0.6
 = Q0.4
 Network 14
 LD M2.4
 = Q0.5
 2 / 7
 task5 / MAIN (OB1)
 Network 15
 LD M0.0
 O M0.3
 AN M0.1
 = M0.3
 Network 16
 LD M0.3
 AN M0.1
 S Q1.4, 1
 Network 17
 LD I0.7
 O T49
 O T52
 O T51
 O T46
 R Q1.4, 1
 Network 18
 LD M0.3
 TON T41, 50

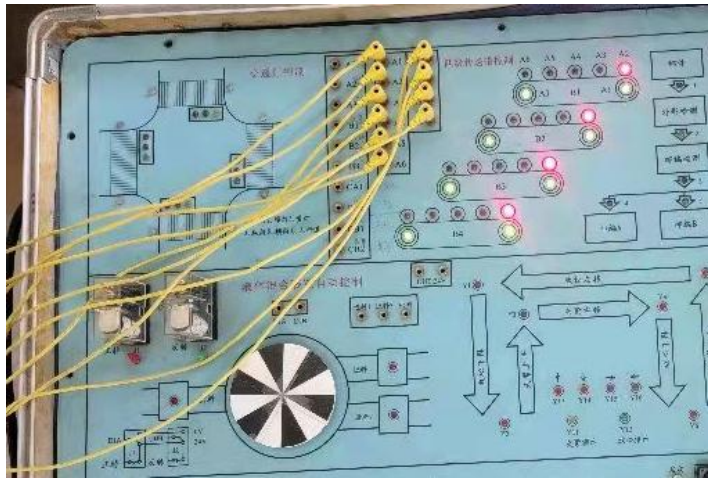


	O T51 O T52 O T46 O I0.7 = M0.6 Network 28 LD T44 TON T45, 50 Network 29 LD T45 TON T46, 50 4 / 7 task5 / MAIN (OB1) Network 30 LD I0.4 TON T47, 50 Network 31 LD T47 TON T48, 50 Network 32 LD T48 TON T49, 50 Network 33 LD I0.5 TON T50, 50 Network 34 LD T50 TON T51, 50 Network 35 LD I0.6 TON T52, 50
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5.5 Experimental results



5.6 Experiment summary

In this project we learn about Control of the four-stage conveyor belt. It's a very important control method in beverage industry. Conveyor systems in the beverage industry are essential for numerous processes on the bottling line: filling, capping, labelling, packing, shipping, and more. A speedy, efficient line not only keeps production on schedule, but can help growing businesses scale to meet demand and more. By learning this project now we have a clear knowledge about the convers system and its ladder diagram.

6 Experiment 6: Water tower water level control

6.1 Purpose of the experiment

1. Master the usage of function instructions
2. Master the design of the water level control program of the water tower

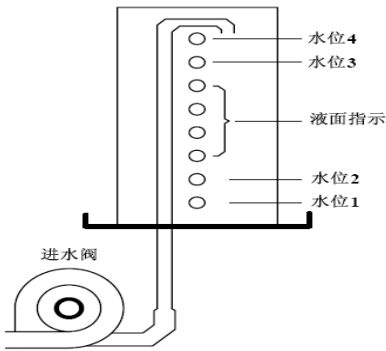
实验器材

- 1、PC 机一台
- 2、PLC 实验箱一台
- 3、水塔水位控制模块
- 4、编程电缆一根
- 5、导线若干

6.2 Experimental content and steps

- 1、Design requirements

Design a water tower water level control program. The module diagram is as follows:



Water tower water level control module schematic

On the control panel, the four green indicators in the water tank position indicate the high water level liquid level sensor and the low water level sensor respectively. The button water level 1, water level 2, water level 3, and water level 4 are used to simulate high and low water level sensors, respectively. Working status. The water level indicator is on, indicating that the current water level is lower than the indicated water level.

The four green lights (3, 4, 5, 6) in the middle are used to indicate the liquid level. The cycle lights up from the bottom to the top, indicating that the water surface is rising.

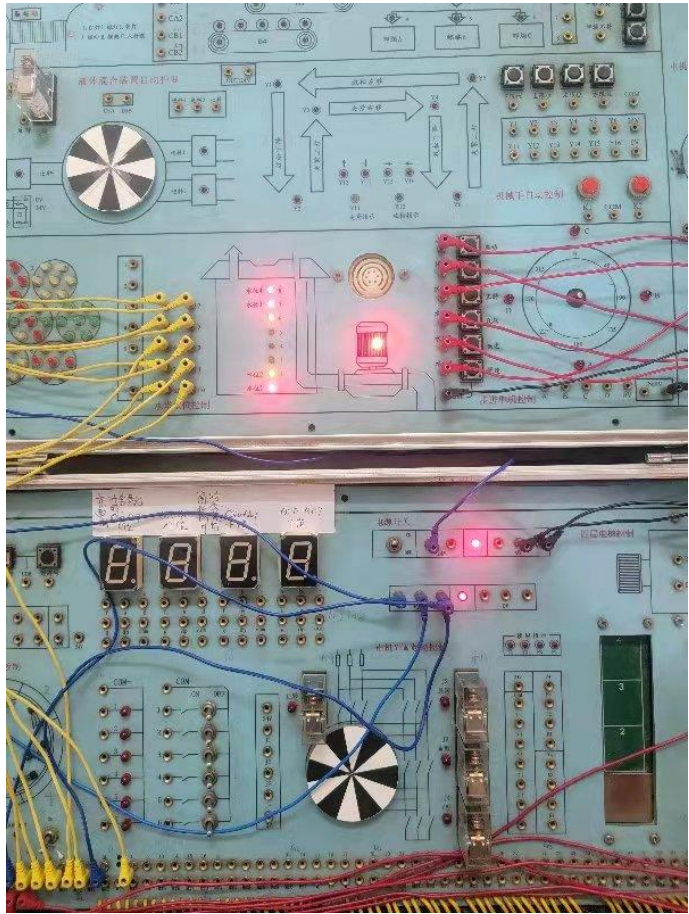
The low water level indicator is on (the high water level indicator is also on), indicating that the water tank is out of water. At this time, the water inlet valve is open (the indicator light is on), the water tank starts to enter the water; press the water level 1 button, the water level 1 indicator is off (the high water level light is still on); press the water level 2 button, the water level 2 indicator is off (high water level light) Still light); press the water level 3 button, the water level 3 indicator is off (the high water level light is still on), then the middle four green lights are slowing down, indicating that the water surface is slowly rising; press the water level 4 button, the water level 4 indicates When the light is off, the middle four green lights are off, indicating that the water tank is full. The inlet valve is closed (the indicator is off).

2. determine the input and output ports, and write a program
3. Compile the program, download it to the memory of the PLC host after error-free, and run the program.
4. debug the program until it meets the design requirements.

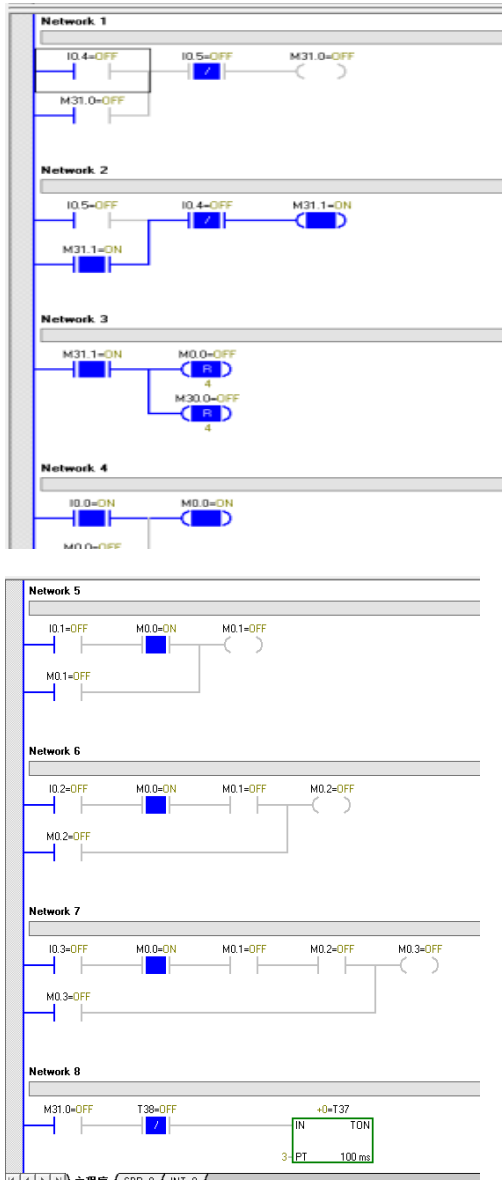
Input			Output		
Host	Experimental module	Comment	Host	Experimental module	Comment
I0.0	Water level 1	Water level 1 button	Q0.0	SP	buzzer
I0.1	Water level 2	Water level 2 button	Q0.1	DJ	Motor work indicator
I0.2	Water level 3	Water level 3 button	Q0.2	Indicator light 1	Water level 1
I0.3	Water level 4	Water level 4 button	Q0.3	Indicator light 2	Water level 2
I0.4	Start	Start button	1L	24V	
I0.5	Stop	Stop button			

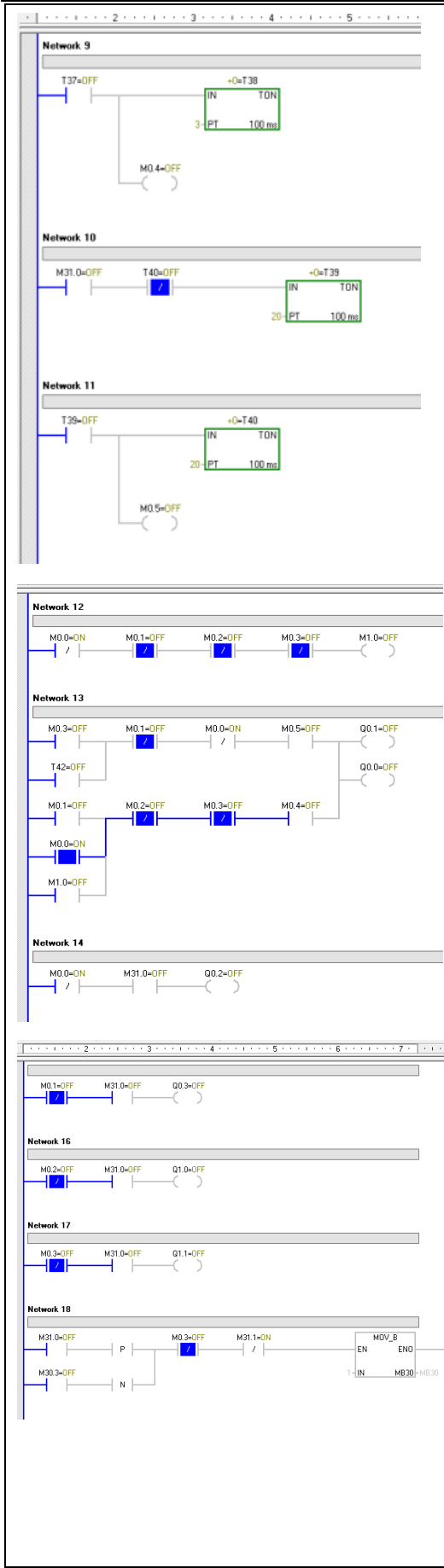
1M	24V		Q0.4	Indicator light 3	3, 4, 5, 6 combination indicates liquid level change
	0V←→COM		Q0.5	Indicator light 4	
			Q0.6	Indicator light 5	
			Q0.7	Indicator light 6	
			2L	24V	
			Q1.0	Indicator light 7	Water level 3 button
			Q1.1	Indicator light 8	Water level 4 button
			3L	24V	

6.3 Experimental wiring

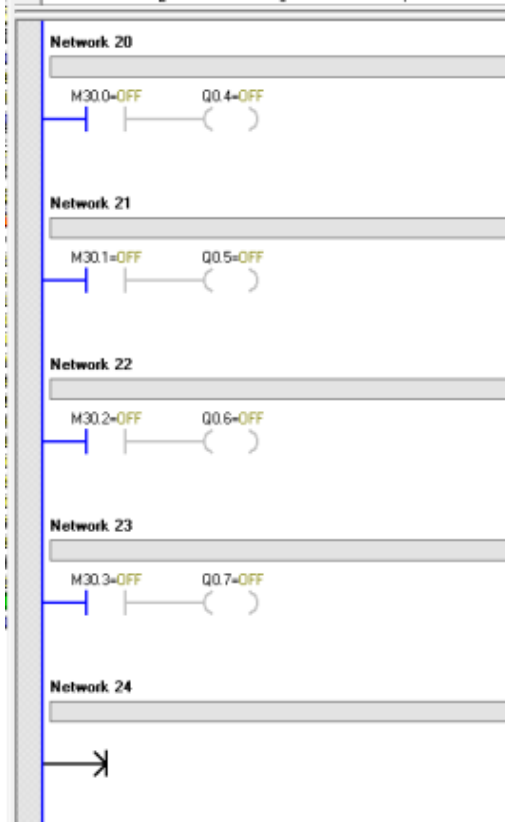


6.4 Experimental program

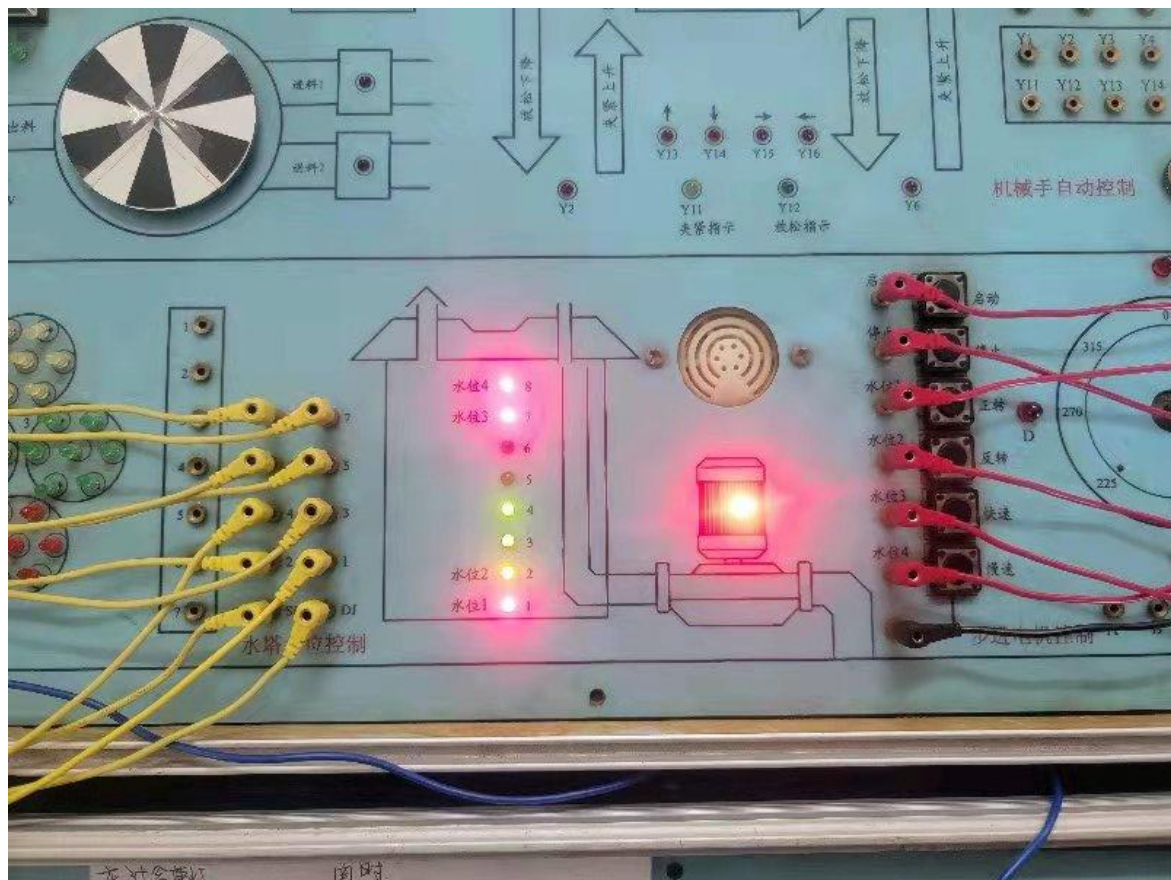
LADDER	STL
	Network 1 LD I0.4 O M31.0 AN I0.5 = M31.0 Network 2 LD I0.5 O M31.1 AN I0.4 = M31.1 Network 3 LD M31.1 R M0.0, 4 R M30.0, 4 Network 4 LD I0.0 O M0.0 = M0.0 Network 5 LD I0.1 A M0.0 O M0.1 = M0.1 Network 6 LD I0.2 A M0.0 A M0.1 O M0.2 = M0.2 1 / 6 task6 / $\ddot{\text{O}}^{\div 3} \dot{\text{I}} \ddot{\text{O}}$ (OB1) Network 7 LD I0.3 A M0.0 A M0.1 A M0.2 O M0.3 = M0.3 Network 8 LD M31.0



AN T38
TON T37, 3
Network 9
LD T37
TON T38, 3
= M0.4
Network 10
LD M31.0
AN T40
TON T39, 20
Network 11
LD T39
TON T40, 20
= M0.5
Network 12
LDN M0.0
AN M0.1
AN M0.2
AN M0.3
= M1.0
2 / 6
task6 / $\ddot{O} \div^3 \dot{I} \dot{D} \dot{o}$ (OB1)
Network 13
LD M0.3
O T42
AN M0.1
AN M0.0
A M0.5
LD M0.1
O M0.0
O M1.0
AN M0.2
AN M0.3
A M0.4
OLD
= Q0.1
= Q0.0
Network 14
LDN M0.0
A M31.0
= Q0.2
Network 15
LDN M0.1
A M31.0

	= Q0.3 Network 16 LDN M0.2 A M31.0 = Q1.0 Network 17 LDN M0.3 A M31.0 = Q1.1 Network 18 LD M31.0 EU LD M30.3 ED OLD AN M0.3 AN M31.1 MOVB 1, MB30 3 / 6 task6 / Ö÷³İĐò (OB1) Network 19 LD M0.4 EU SLB MB30, 1 Network 20 LD M30.0 = Q0.4 Network 21 LD M30.1 = Q0.5 Network 22 LD M30.2 = Q0.6 Network 23 LD M30.3 = Q0.7 4 /
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6.5 Experimental results



6.6 Experiment summary

This is our last project in PLC experiment course. In this project we learn about Water tower water level control. The operating statuses of these switches are listed here:

- ✓ Start pushbutton (normally-open)
- ✓ Stop pushbutton (normally-closed)
- ✓ Low water level indicator (normally-closed)
- ✓ high water level indicator (normally-open)
- ✓ over flow level indicator

working on a project with a topic that is relevant to our departments. First and foremost, I am grateful to have excellent guidance from NINA laoshi, as well as a fantastic team. firstly, my greatest success is gaining a wide range of information on many topics. Normally, we obtain knowledge via books, but by working on a project like this, we can do so practically. To execute our assignment, we must read various books and reference paper. We worked with a variety of applications and received a lot of support from dozens of websites, which helped us to deepen our knowledge. working on this project, I learned a lot about PLC. Finally, I am grateful to my teacher for providing us for the opportunity to experience more and more.